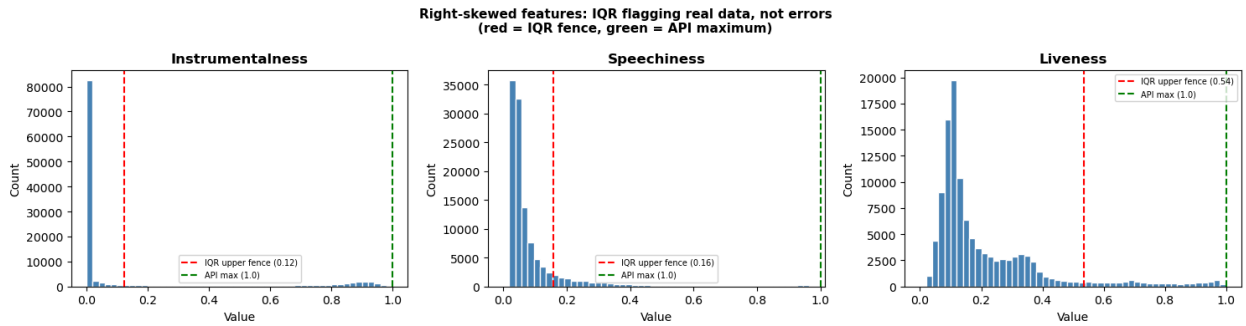
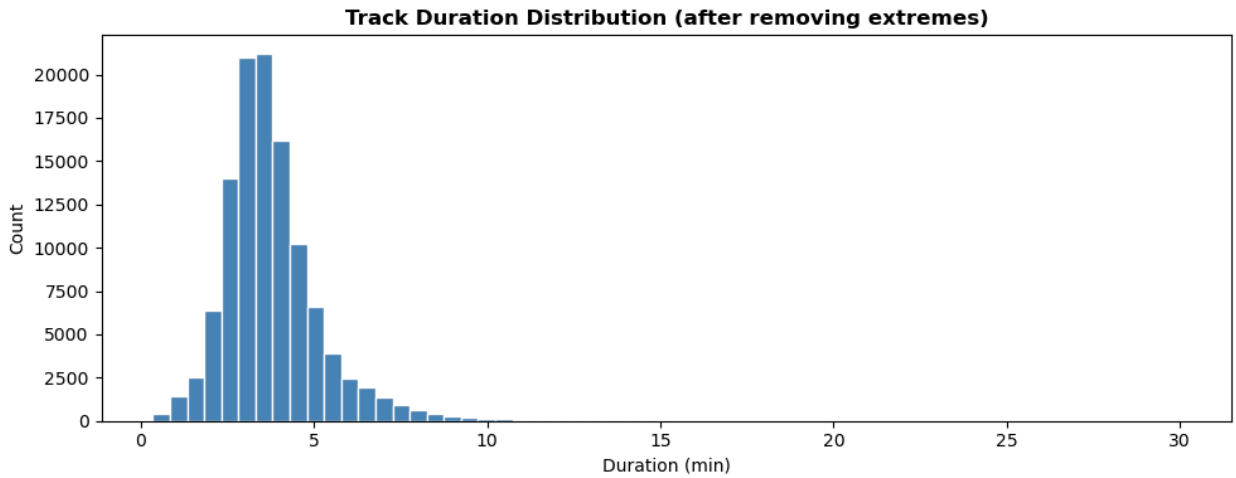
Final Presentation Visuals

Preprocessing





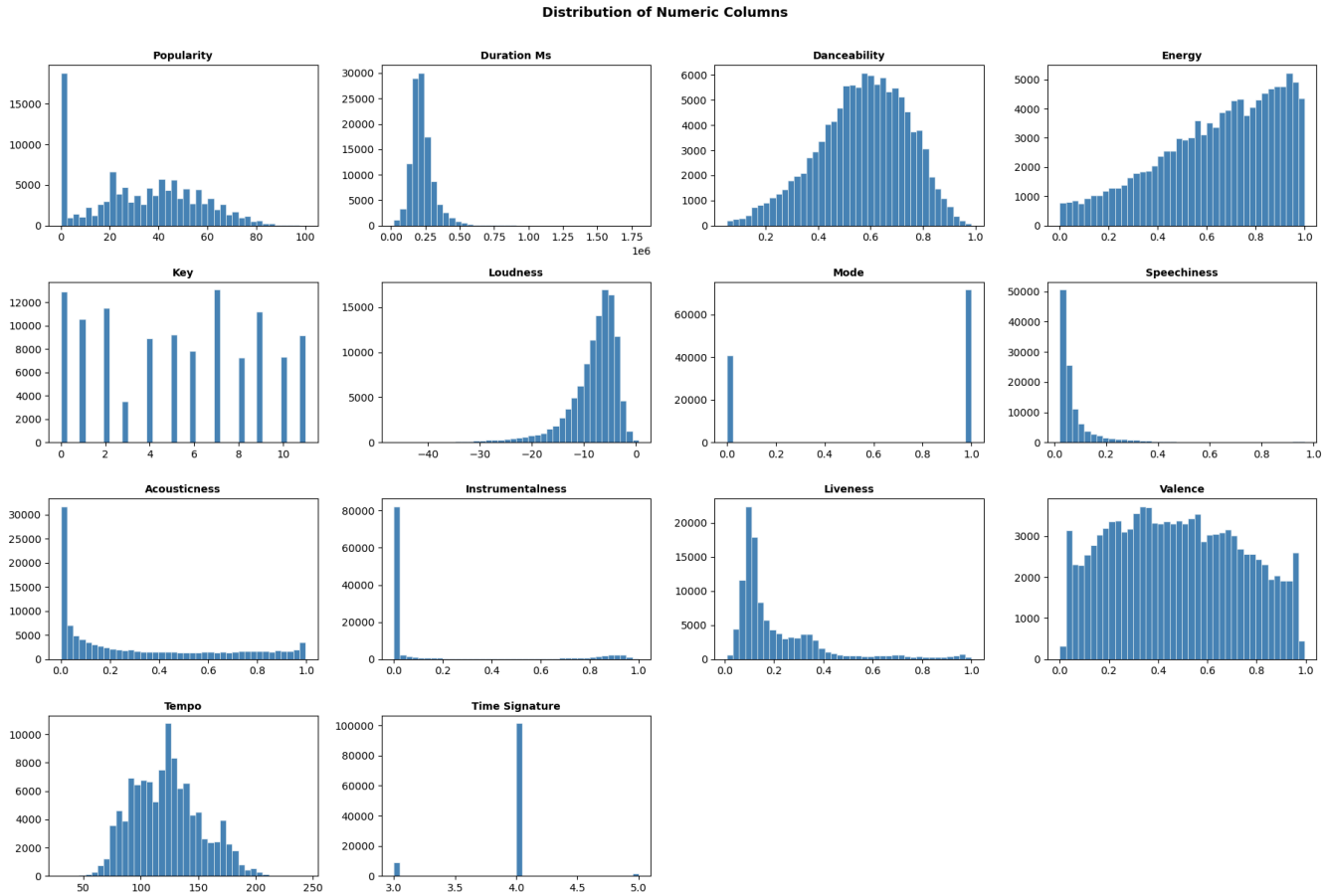

```
Tracks under 20 seconds  : 3
Tracks over 30 minutes  : 34

Shortest tracks (sample):
```

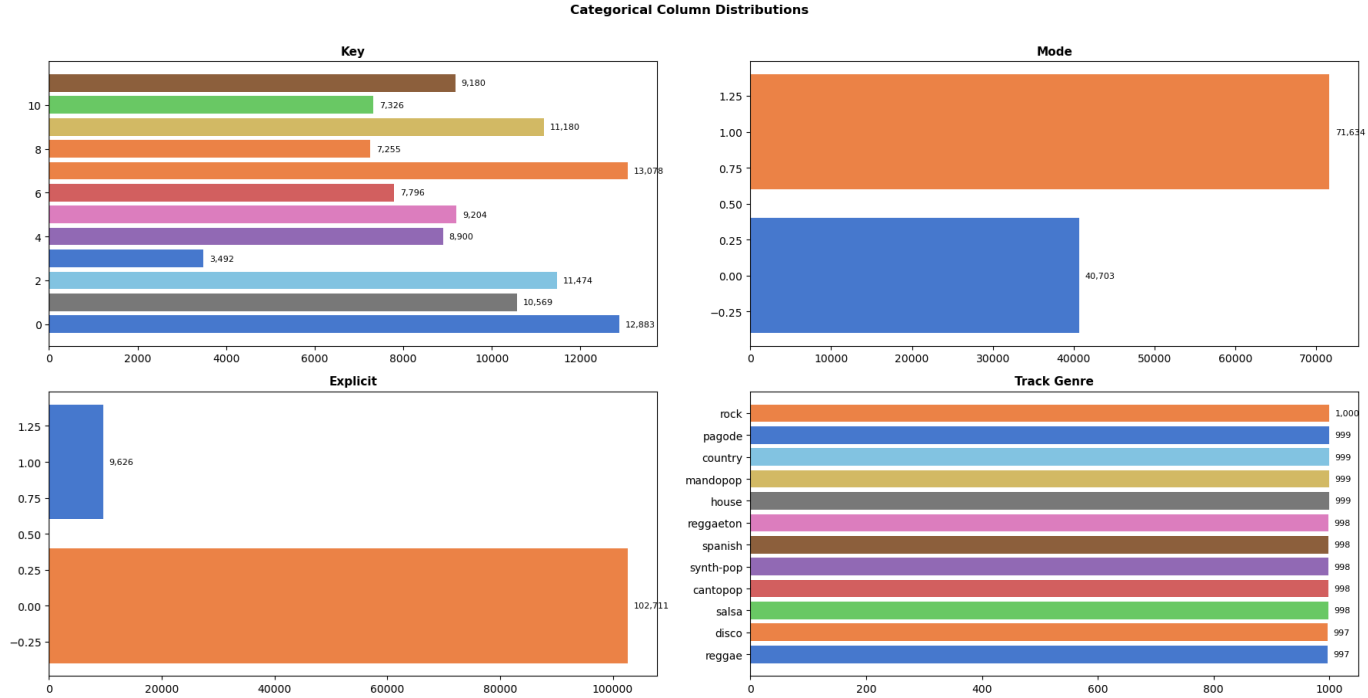
	track_name	artists	duration_ms	track_genre
65900	NaN	NaN	0	k-pop
59775	Screams for a Finale! (feat. Leila's Opera Class)	Leila Bela;Leila's Opera Class	15800	iranian
16856	Andante in C Major, K. 1a	Wolfgang Amadeus Mozart;Ingrid Haebler	17453	classical
59306	V-7	Leila Bela	21120	iranian
59609	Shatter	Leila Bela	21240	iranian

```
Longest tracks (sample):
```

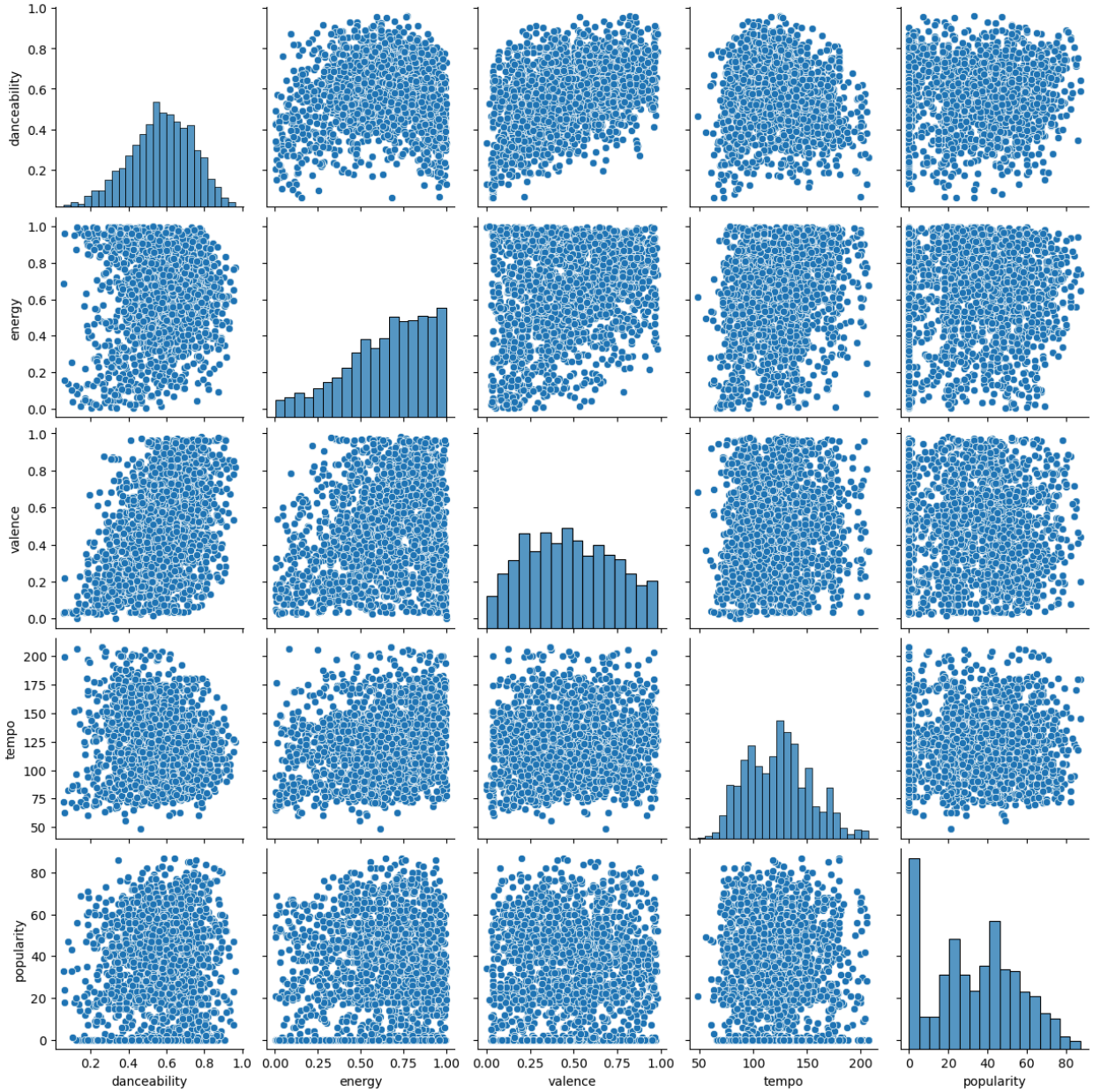
	track_name	artists	duration_ms	track_genre
73617	Unity (Voyage Mix) Pt. 1	Tale Of Us	5237295	minimal-techno
10935	Crossing Wires 002 - Continuous DJ Mix	Timo Maas	4789026	breakbeat
10984	Crossing Wires 002 - Continuous DJ Mix	Timo Maas	4789026	breakbeat
24348	The Lab 03 - Continuous DJ Mix Part 1	Seth Troxler	4730302	detroit-techno
73840	Amnesia Ibiza Underground 10 DJ Mix	Loco Dice	4563897	minimal-techno

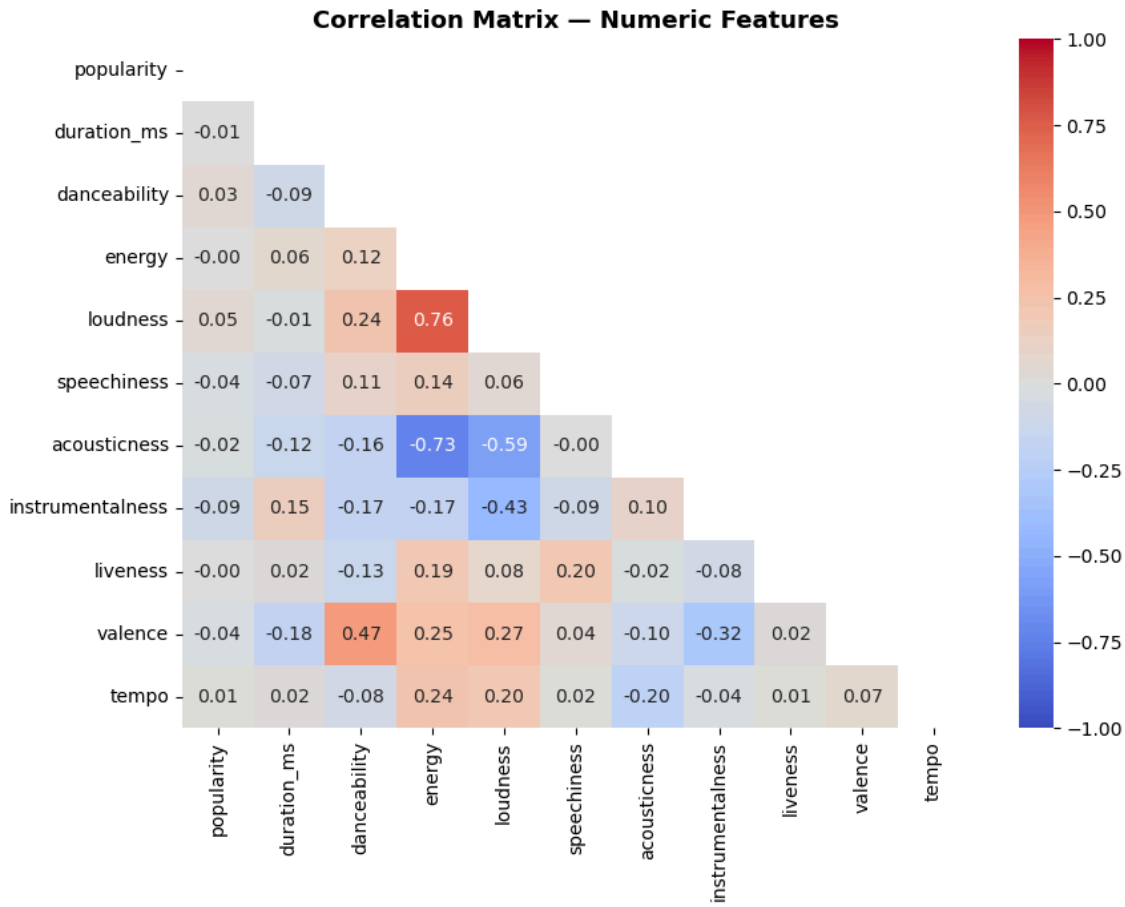
Exploratory Visuals

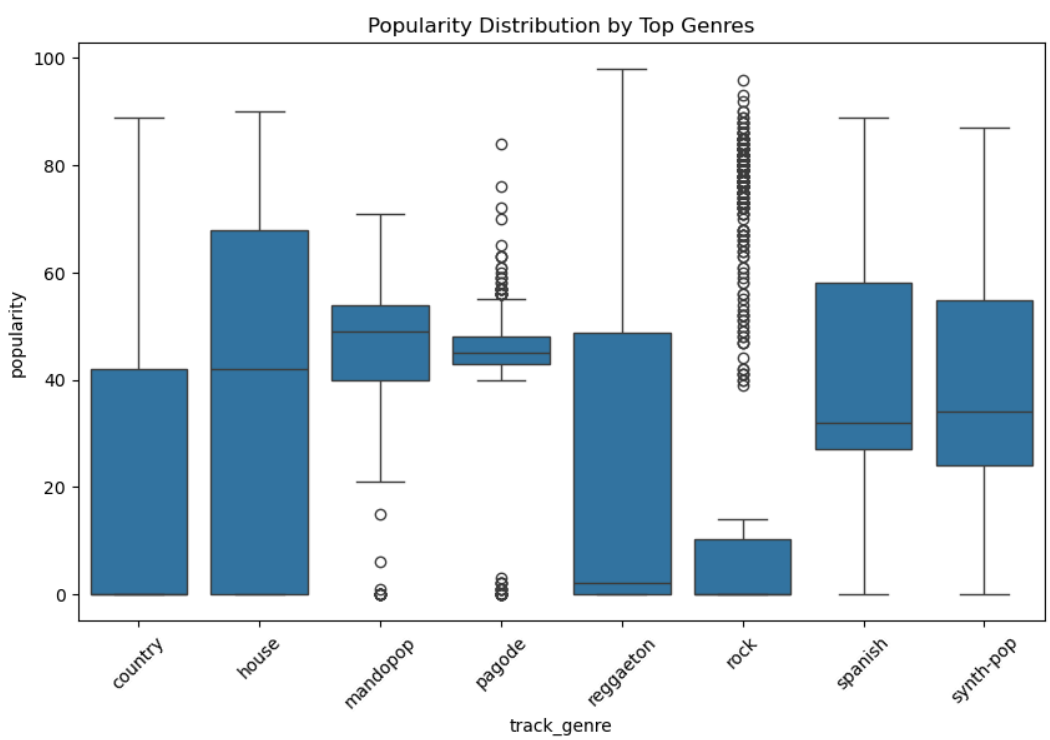


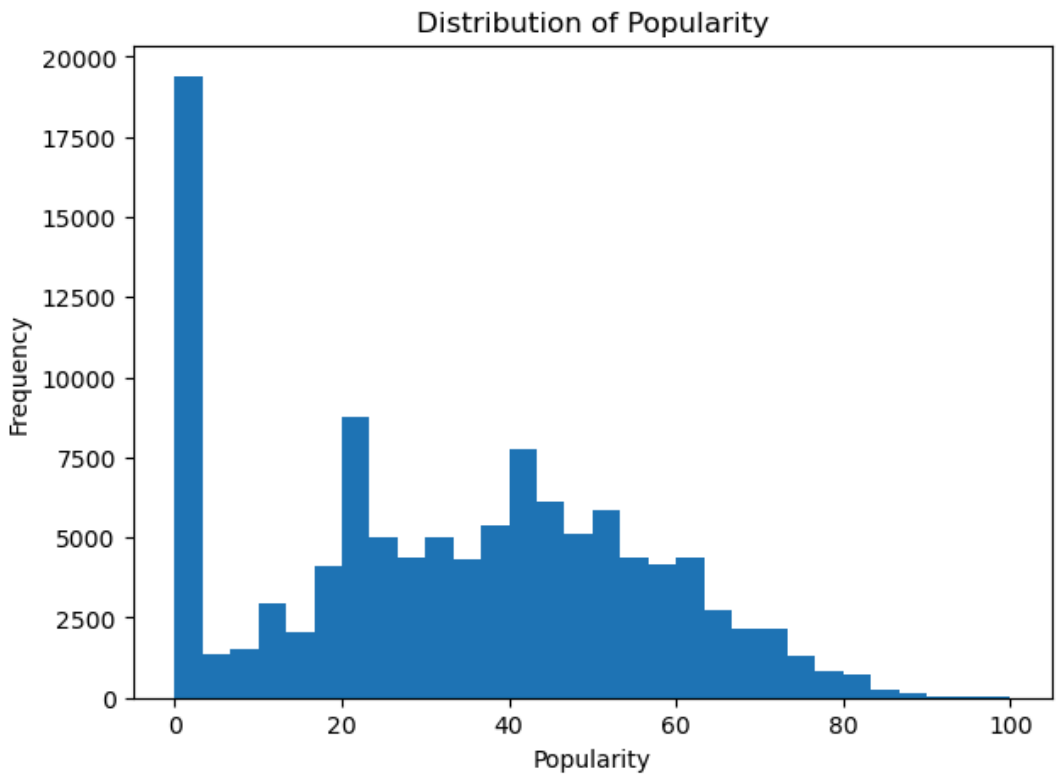


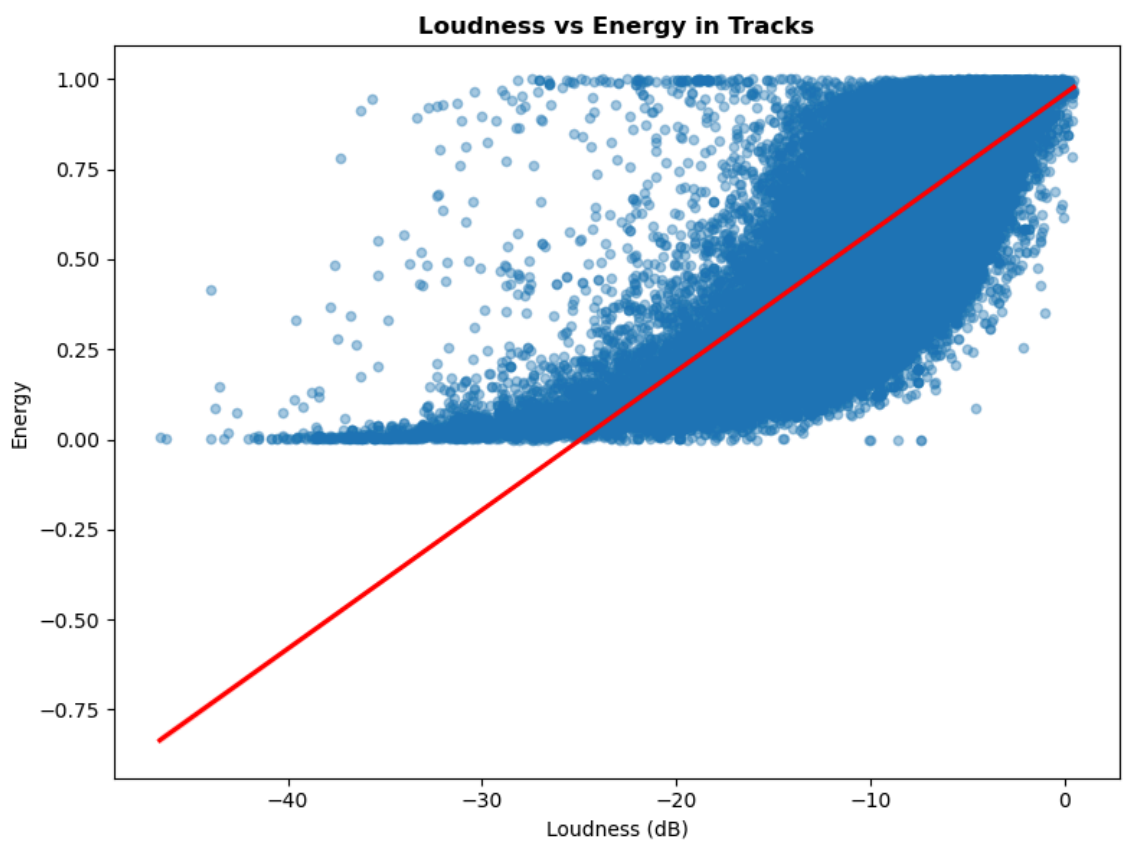
Pairwise Relationships of Key Audio Features




Correlation Matrix — Numeric Features
popularity
duration_ms
danceability
energy
loudness
speechiness
acousticness
instrumentalness
liveness
valence
tempo
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Popularity Distribution by Top Genres
popularity
track_genre
country
house
mandopop
pagode
reggaeton
rock
spanish
synth-pop

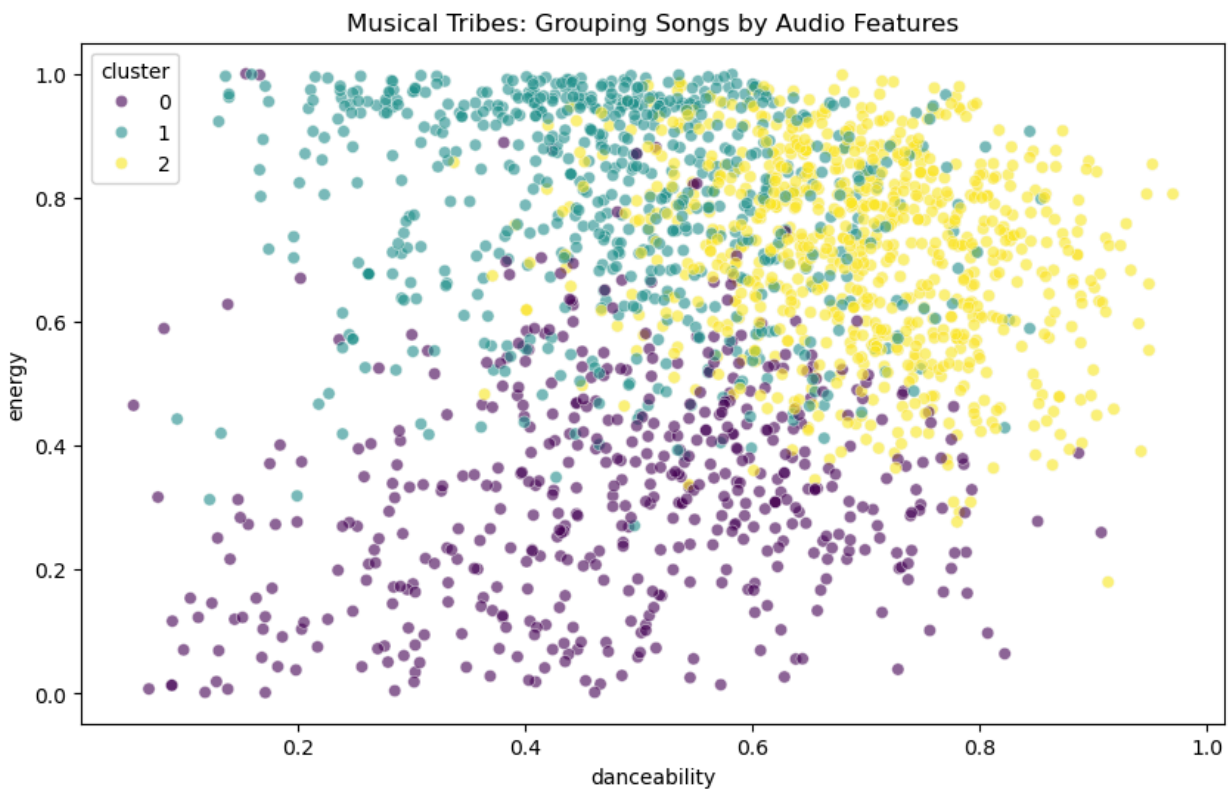

Distribution of Popularity
Frequency
Popularity
0
2500
5000
7500
10000
12500
15000
17500
20000
0
20
40
60
80
100


Loudness vs Energy in Tracks
Energy
Loudness (dB)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−40
−30
−20
−10
0

Overall Analysis

```
VIF Results (Ideally < 5-10):
             feature        VIF
0      danceability  12.446079
1            energy  15.268944
2          loudness   7.164117
3       speechiness   1.807298
4      acousticness   3.568545
5  instrumentalness   1.761088
6          liveness   2.574076
7           valence   6.541496
8             tempo  15.462227
                            OLS Regression Results
==============================================================================
Dep. Variable:             popularity   R-squared:                       0.023
Model:                            OLS   Adj. R-squared:                  0.023
Method:                 Least Squares   F-statistic:                     232.1
Date:                Wed, 22 Apr 2026   Prob (F-statistic):               0.00
Time:                        17:23:58   Log-Likelihood:            -4.0559e+05
No. Observations:               89869   AIC:                         8.112e+05
Df Residuals:                   89859   BIC:                         8.113e+05
Df Model:                           9
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const               36.1465      0.754     47.923      0.000      34.668      37.625
danceability         9.3450      0.523     17.865      0.000       8.320      10.370
energy              -2.9525      0.610     -4.837      0.000      -4.149      -1.756
loudness             0.1273      0.027      4.712      0.000       0.074       0.180
speechiness        -12.2454      0.746    -16.408      0.000     -13.708     -10.783
acousticness        -0.8423      0.346     -2.435      0.015      -1.520      -0.164
instrumentalness    -8.3531      0.289    -28.875      0.000      -8.920      -7.786
liveness             1.1074      0.415      2.670      0.008       0.294       1.920
valence             -9.8908      0.353    -28.007      0.000     -10.583      -9.199
tempo                0.0157      0.003      6.072      0.000       0.011       0.021
==============================================================================
Omnibus:                     9800.118   Durbin-Watson:                   1.998
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2723.817
Skew:                          -0.019   Prob(JB):                         0.00
Kurtosis:                       2.148   Cond. No.                     1.71e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.71e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

```
Predicting High Popularity (Threshold: 35.0):
- Accuracy: 0.5660494926117144
- ROC-AUC Score: 0.5831511533535009

Confusion Matrix:
 [[7006 4530]
 [5220 5712]]

Classification Report:
              precision    recall  f1-score   support

           0       0.57      0.61      0.59     11536
           1       0.56      0.52      0.54     10932

    accuracy                           0.57     22468
   macro avg       0.57      0.56      0.56     22468
weighted avg       0.57      0.57      0.57     22468
```

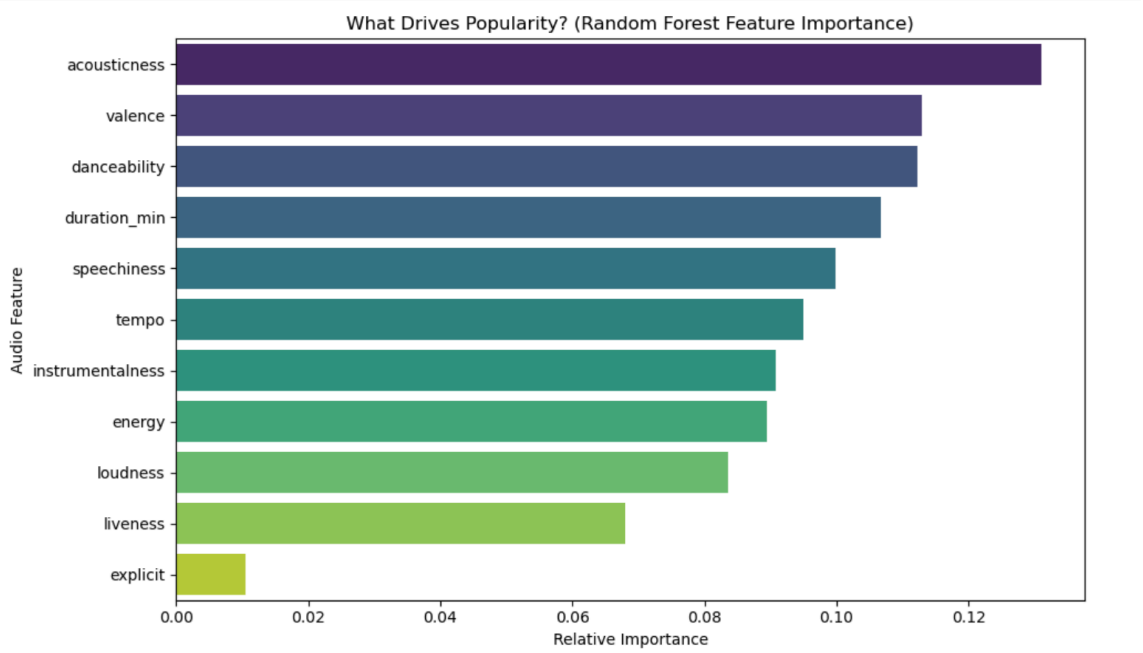


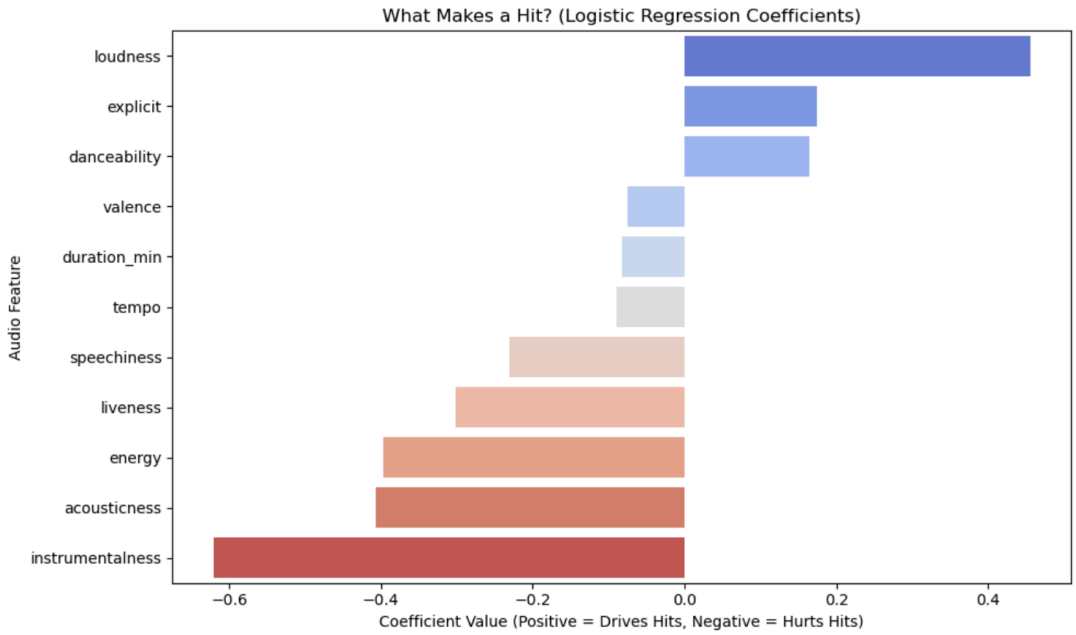

```
Average Feature Values per Cluster:
         danceability    energy  acousticness   valence
cluster
0            0.478479  0.322768      0.763040  0.327181
1            0.483761  0.796214      0.074717  0.315858
2            0.693771  0.713924      0.232759  0.697386
```

Musical Tribes Breakdown

- Cluster 0 | High Acousticness (0.76), Lowest Energy | "The Unplugged / Chill" (Think: Folk, Classical, or Acoustic covers)
- Cluster 1 | Highest Energy (0.80), Lowest Valence & Acousticness | "The High-Energy / Darker" (Think: Metal, Hard Rock, or Techno)
- Cluster 2 | Highest Danceability (0.69) & Valence (0.69) | "The Feel-Good Hits" (Think: Pop, Funk, or Disco)

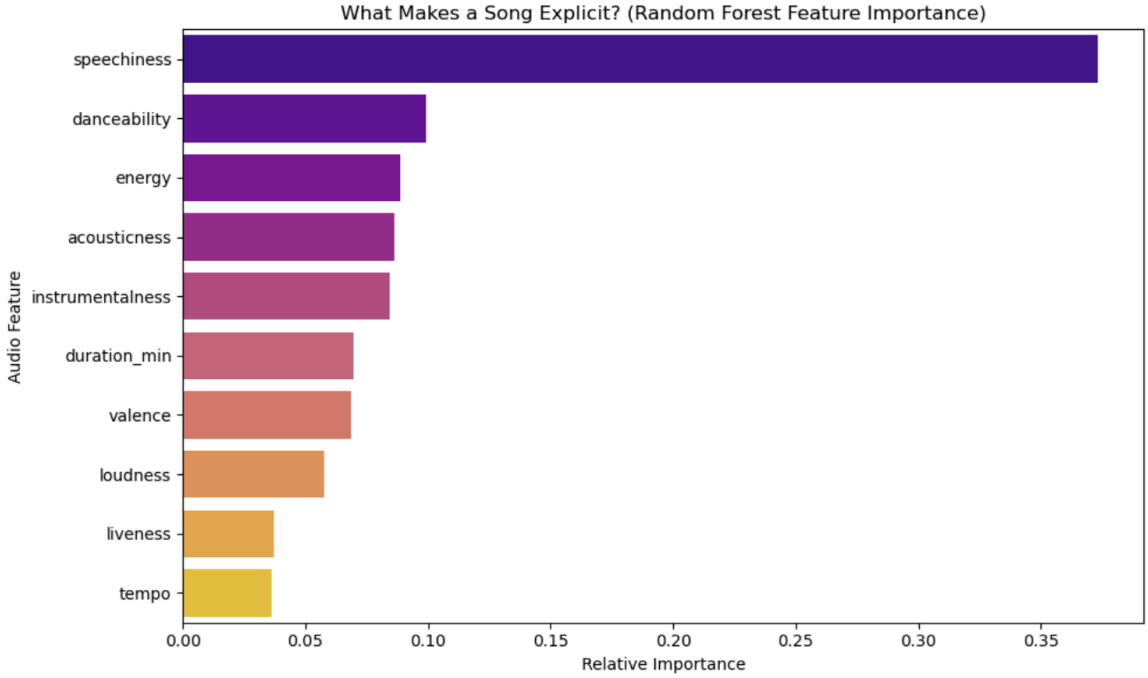
Predictions



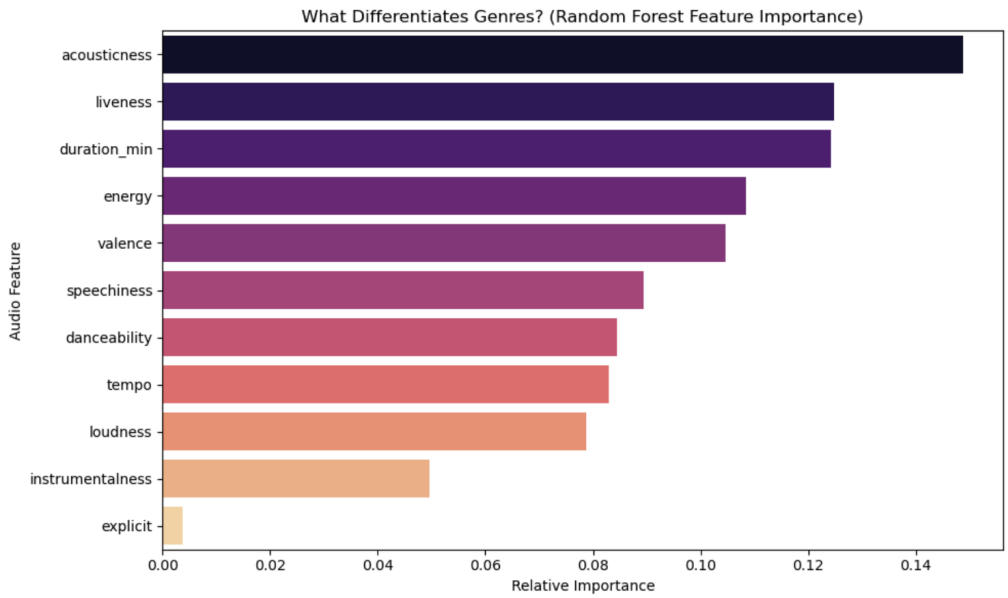


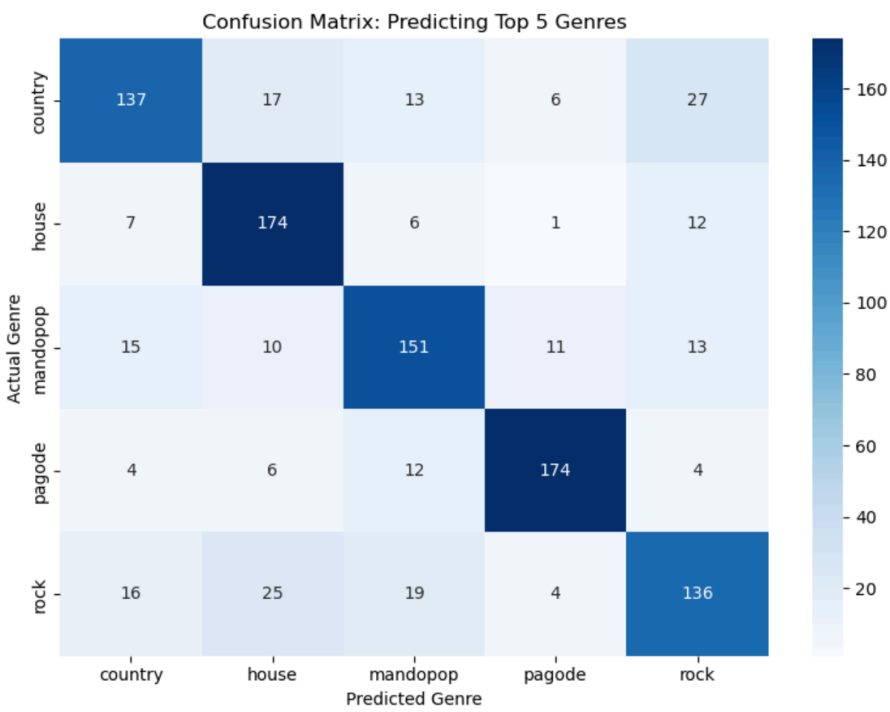

```
Hit Prediction Accuracy: 0.5755296421577355
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.57      0.72     21422
           1       0.08      0.75      0.14      1046

    accuracy                           0.58     22468
   macro avg       0.53      0.66      0.43     22468
weighted avg       0.94      0.58      0.69     22468
```

```
Explicit Prediction Accuracy: 0.8053676339683105
Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.81      0.88     20543
           1       0.27      0.76      0.40      1925

    accuracy                           0.81     22468
   macro avg       0.62      0.79      0.64     22468
weighted avg       0.91      0.81      0.84     22468
```



```
Genre Prediction Accuracy: 0.772
Classification Report:
              precision    recall  f1-score   support

     country       0.77      0.69      0.72       200
       house       0.75      0.87      0.81       200
    mandopop       0.75      0.76      0.75       200
      pagode       0.89      0.87      0.88       200
        rock       0.71      0.68      0.69       200

    accuracy                           0.77      1000
   macro avg       0.77      0.77      0.77      1000
weighted avg       0.77      0.77      0.77      1000
```



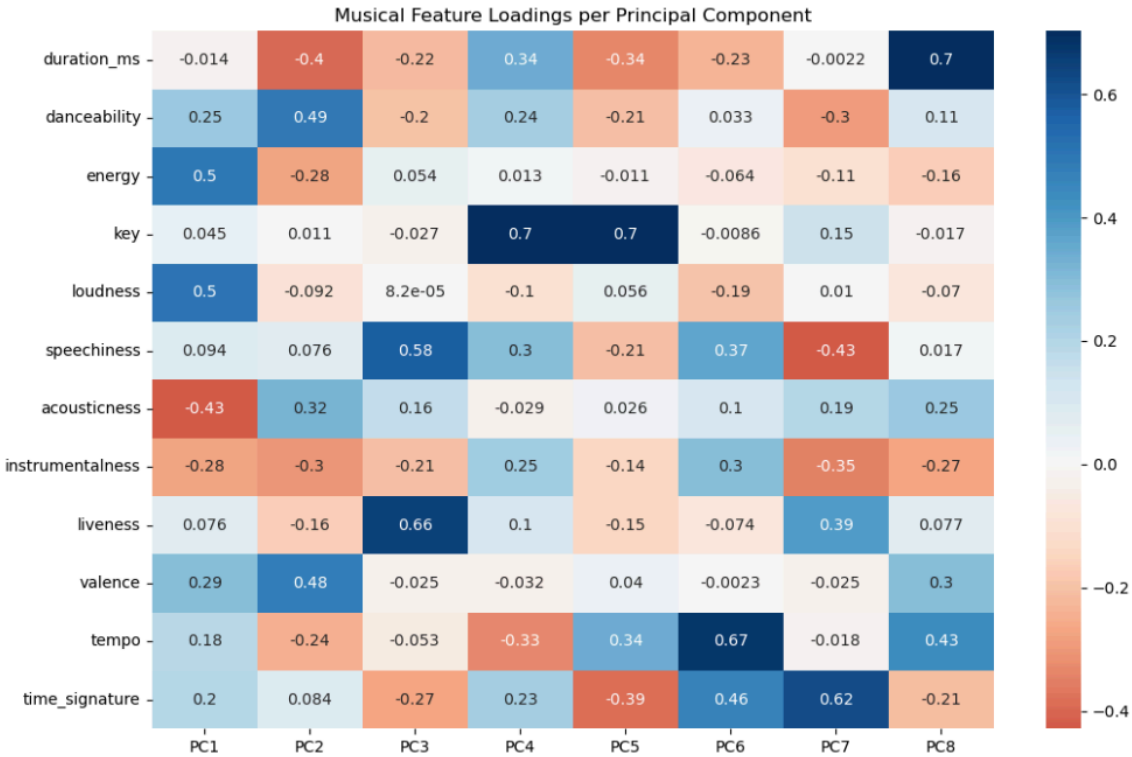


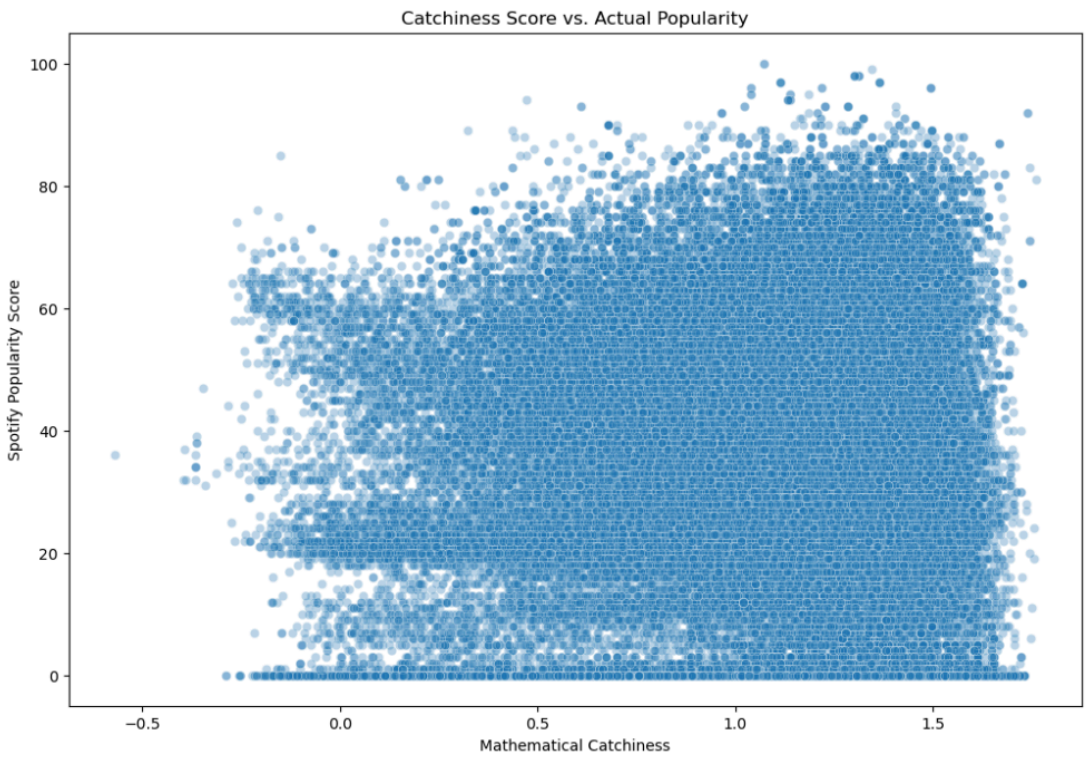
Catchiness Analysis

```
Components needed for 80% variance: 8
Variance retained with new k: 81.1%
```

```
--- Final Model Performance ---
Split Accuracy    AUC
Train   68.20% 0.8532
  Val   54.75% 0.7363
 Test   54.77% 0.7369
```

```
PC2     0.138547
PC1     0.136761
PC8     0.135681
PC4     0.129502
PC3     0.124398
PC6     0.114815
PC7     0.114335
PC5     0.105960
```


Musical Feature Loadings per Principal Component
duration_ms -0.014 -0.4 -0.22 0.34 -0.34 -0.23 -0.0022 0.7
danceability 0.25 0.49 -0.2 0.24 -0.21 0.033 -0.3 0.11
energy 0.5 -0.28 0.054 0.013 -0.011 -0.064 -0.11 -0.16
key 0.045 0.011 -0.027 0.7 0.7 -0.0086 0.15 -0.017
loudness 0.5 -0.092 8.2e-05 -0.1 0.056 -0.19 0.01 -0.07
speechiness 0.094 0.076 0.58 0.3 -0.21 0.37 -0.43 0.017
acousticness -0.43 0.32 0.16 -0.029 0.026 0.1 0.19 0.25
instrumentalness -0.28 -0.3 -0.21 0.25 -0.14 0.3 -0.35 -0.27
liveness 0.076 -0.16 0.66 0.1 -0.15 -0.074 0.39 0.077
valence 0.29 0.48 -0.025 -0.032 0.04 -0.0023 -0.025 0.3
tempo 0.18 -0.24 -0.053 -0.33 0.34 0.67 -0.018 0.43
time_signature 0.2 0.084 -0.27 0.23 -0.39 0.46 0.62 -0.21
PC1 PC2 PC3 PC4 PC5 PC6 PC7 PC8
0.6
0.4
0.2
0.0
−0.2
−0.4


Catchiness Score vs. Actual Popularity
Spotify Popularity Score
100
80
60
40
20
0
−0.5
0.0
0.5
1.0
1.5
Mathematical Catchiness